DHARANENDRA CHENCHALA

Machine Learning Engineer | AI/ML Undergraduate

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PROFESSIONAL SUMMARY

Results-driven AI/ML undergraduate with hands-on experience building and deploying machine learning, deep learning, and NLP systems. Proficient in Python, TensorFlow, and PyTorch with applied experience across computer vision, natural language processing, and bioinformatics. Demonstrates consistent ability to translate complex datasets into accurate, scalable models — achieving 85–94% accuracy across multiple independent projects. Actively developing expertise in transformer architectures and large language model (LLM) integration. Seeking to contribute strong fundamentals and a rapid learning trajectory to a high-impact ML engineering team.

TECHNICAL SKILLS

Languages: Python, C, C++, Java

ML Frameworks: TensorFlow, PyTorch, Scikit-learn, Keras

NLP & CV: Tokenization, Vectorization, Intent Classification, Image Classification, Data Augmentation, OpenCV

Libraries: NumPy, Pandas, Matplotlib, OpenCV, NLTK

Tools & Platforms: Git, GitHub, Docker, MySQL, Google Colab, Jupyter Notebook

Core ML Concepts: CNNs, Supervised Learning, Feature Engineering, Hyperparameter Tuning, Model Optimization, Data Preprocessing, Transfer Learning

Domains: Computer Vision, Natural Language Processing (NLP), Bioinformatics, Deep Learning

PROJECTS & TECHNICAL EXPERIENCE

Crop Disease Detection — Computer Vision | *Personal Project* *Jan 2025 – Present*

CNN · PyTorch · Image Classification · Data Augmentation · PlantVillage Dataset

- Architected a CNN image classifier trained on 20,000+ PlantVillage images, achieving **94% validation accuracy** across 30+ plant disease categories.
- Engineered an augmentation pipeline (rotation, flipping, zoom normalization) that improved model accuracy by **15%** and reduced overfitting on imbalanced class distributions.
- Reduced misclassification rate by iterative hyperparameter tuning (learning rate, dropout rate, batch size), lifting baseline accuracy from **~79% to 94%** across 30+ disease classes.
- Designed a prototype early-warning inference system for field-level agricultural disease detection, targeting scalable low-compute deployment.

AI-Driven DNA Sequence to Structural Modeling | *Personal Project* *Jan 2025 – Apr 2025*

Deep Learning · Bioinformatics · Genomic Sequence Analysis · Python

- Designed a deep learning model to predict sequence-to-structure relationships across 10,000+ genomic sequences, achieving **~85% prediction accuracy**.
- Built optimized bioinformatics preprocessing pipelines that reduced data preparation time by **30%**, enabling faster model iteration and experimentation cycles.
- Applied domain-specific feature engineering to genomic data, improving structural prediction quality and reducing computational overhead in sequence analysis tasks.
- Integrated bioinformatics domain knowledge with deep learning to enhance biological insight extraction from raw DNA sequence data.

AI Chatbot — Natural Language Processing | *Personal Project* *Aug 2024 – Jan 2025*

NLP · Intent Classification · Tokenization · Vectorization · Python

- Built an intent classification chatbot trained on 1,000+ labeled queries across 15+ intent categories, achieving **89% classification accuracy** on held-out test data.
- Reduced average query response latency by **40%** by replacing baseline tokenization with optimized vectorization pipelines, enabling near real-time user interaction.
- Automated end-to-end query handling across multiple domains, improving response consistency and reducing unrecognized input fallback rate.

- Designed and validated training data across varied query structures, improving model robustness to real-world phrasing variation.

NLP Text Analysis System — Sentiment Analysis | *Personal Project* *Jun 2024 – Aug 2024*
Machine Learning · TF-IDF · Text Classification · Feature Engineering · Python

- Developed a multi-class sentiment classifier on 10,000+ text samples, achieving **91% accuracy** — a 12% improvement over the unprocessed baseline model.
- Improved feature representation using TF-IDF weighting and custom feature engineering, directly driving the **12-point accuracy gain** over raw text input.
- Built a reusable preprocessing pipeline handling noise removal, tokenization, and vectorization, enabling structured insight extraction from unstructured user-generated text.

EDUCATION

B.Tech — Computer Science Engineering (AI & ML) | *Sir Padampat Singhania University* *2023 – 2027*
Relevant Coursework: Machine Learning, Data Structures & Algorithms, Database Management Systems, Deep Learning, Computer Vision

Intermediate (Class XI–XII) | Government Junior College, Nalgonda, Telangana | 2023
Secondary (Class X) | SPR School, Miryalaguda, Telangana | 2021

CERTIFICATIONS

GenAI Powered Data Analytics Job Simulation — Forage | 2025
Explore Machine Learning using Python — 2024
SQL and Relational Databases 101 — IBM | 2024